

KARTHIK NAIK

Software Development Engineer | Java Backend Engineer

Bangalore, India | +91-78294 35689 | karthiknknai297@gmail.com

LinkedIn: karthik-naik- | GitHub: KaRtHiK-030 | Portfolio: karthik-030.github.io

PROFESSIONAL SUMMARY

Computer Science Engineering graduate with hands-on experience building backend applications using Java and Spring Boot. Skilled in developing RESTful APIs, concurrent systems, and database-driven applications using MySQL and Docker. Strong foundation in Data Structures & Algorithms, Object-Oriented Programming, Multithreading, and System Design, with experience building production-ready software projects.

TECHNICAL SKILLS

Programming Languages:	Java, Kotlin, Python, SQL
Backend Technologies:	Spring Boot, REST APIs, Multithreading, Socket Programming
Web Technologies:	HTML, CSS
Databases:	MySQL, Firebase (Firestore / Realtime DB)
ORM:	Hibernate, Spring Data JPA
Core CS:	Data Structures & Algorithms, OOP, Concurrency, System Design, Caching, MVVM Architecture
Tools & DevOps:	Git, GitHub, Docker, Gradle, Maven, Postman, Swagger/OpenAPI
Testing & Quality:	JUnit, Integration Testing, JaCoCo

EXPERIENCE

MindMatrix	Bangalore, India
<i>Android App Development Intern – Generative AI</i>	Feb 2026 – May 2026
- Developed Android applications integrating AI capabilities and REST APIs to improve user experience and application functionality. - Built and integrated REST APIs to support scalable frontend-backend communication and faster response times. - Collaborated in an Agile development environment with Git-based version control, code reviews, and performance optimization.	
Samsung Innovation Campus	Bangalore, India
<i>IoT Trainee</i>	Sep 2024 – Dec 2024
- Developed IoT solutions for real-time monitoring using sensors, microcontrollers, and communication protocols. - Implemented hardware-software integration workflows for efficient data collection and automation.	

PROJECTS

URL Shortener Service	Java, Spring Boot, Spring Data JPA, MySQL, Docker, Swagger/OpenAPI
- Built and implemented a RESTful URL shortening service using Java, Spring Boot, Spring Data JPA, and MySQL, with Base62 encoding for compact URL generation. - Developed RESTful APIs supporting custom aliases, URL expiration, QR code generation, and click analytics. - Documented REST APIs using Swagger/OpenAPI, implemented unit and integration tests with JUnit, measured code coverage using JaCoCo, and containerized the application with Docker. - Handled application errors gracefully with global exception handling for a robust API experience.	
Multithreaded HTTP Proxy Server	Java, Socket Programming, ExecutorService, LRU Cache, Rate Limiting
- Built and implemented a multithreaded HTTP proxy server in Java to handle and forward client requests using socket programming. - Designed an LRU cache with TTL-based eviction to reduce redundant upstream requests. - Implemented a sliding window rate limiter to control traffic and prevent abuse. - Used ExecutorService to efficiently manage concurrent client connections. - Exposed a metrics endpoint for monitoring server performance and benchmarked the proxy at ~1150 requests/sec.	
NammaKelsa	Kotlin, Jetpack Compose, Firebase, MVVM, Firestore
- Built and implemented a two-sided Android marketplace application connecting skilled workers with hirers using Kotlin and Jetpack Compose. - Solved real-time communication and request-tracking needs between users on a shared mobile platform. - Implemented MVVM architecture with Repository pattern using Firebase Authentication, Cloud Firestore, and Realtime Database-powered chat. - Implemented worker and hirer workflows with request management, reviews, ratings, favorites, and profile management backed by Firebase Storage.	

EDUCATION

Cambridge Institute of Technology (VTU)	May 2026
B.E. in Computer Science Engineering	CGPA: 7.21 / 10

CERTIFICATIONS

- NPTEL – Data Structures and Algorithms Using Java (2025)
- Samsung Innovation Campus – Internet of Things (IoT) (2024)